DAY 20 – Machine Learning Algorithms Overview

Day 20 expanded understanding of machine learning by exploring the core types of algorithms used in real-world data science. The session emphasized supervised, unsupervised, and reinforcement learning—giving a broader perspective beyond forecasting.

🔹 1. Supervised Learning

We studied algorithms where the model learns from labeled data:

• Linear Regression
• Logistic Regression
• Decision Trees
• Random Forest
• Support Vector Machines

These are useful for prediction, classification, and trend analysis.

🔸 2. Unsupervised Learning

We explored techniques that detect patterns without labels:

✓ K-Means Clustering
✓ Hierarchical Clustering
✓ PCA (Dimensionality Reduction)

Used for grouping, visualization, and anomaly detection.

🔸 3. Reinforcement Learning Basics

We learned RL through simple examples:

• Agent-environment interaction
• Reward-based learning
• Applications in gaming and robotics

Although not required for this project, RL builds advanced ML knowledge.

🔸 4. Where These Algorithms Are Used

Real-world applications included:

• Spam detection
• Stock market analysis
• Customer segmentation

• Recommendation systems
• Fraud detection

Understanding these helps identify correct algorithms for future tasks.

🔸 5. Importance of Algorithm Selection

Choosing the right algorithm depends on:

• Data type
• Problem category
• Accuracy requirements
• Computational cost
• Model interpretability

Good selection improves performance dramatically.

Key Takeaways from Day 20

• ML algorithms have different purposes
• Supervised learning is most widely used
• Unsupervised learning reveals hidden patterns
• Reinforcement learning is growing rapidly
• Algorithm choice defines project success